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Review

Economical assessment and applications of photovoltaic/thermal hybrid solar technology: A review

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ABSTRACT

In recent years, photovoltaic technology has been developing quickly and studies found that only less than 20% solar energy was converted into electricity while more than 50% of the incident solar radiation is transformed into heat. PV module electrical efficiency and operating temperature can be improved by removing the excess heat. This led to the concept of a hybrid collector called PVT collector. Cooling of PV cells has become an important study in order to increase the PV panel efficiency, improvement in power output and optimum performance parameters. Numerous studies theoretical, experimental and economical assessment haves been carried out for different hybrid PVT solar collectors designs. Other alternative conceptions can be improved in different applications. This paper gives a review of the historical and recent trend of PVT development technology, in particular the performance and economic feasibility of PVT system for different application areas. Future recommendations and critical assessment are made to overcome the barriers and challenges focusing of improving PVT technology.

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1. Introduction

1.1. PVT classification

Solar PV panels are increasingly used all over the world due to their capability to work under diffuse radiation. So, it is important to know how PV panels respond to changed climatic conditions. In practice only 15–20% of the solar irradiation can be transformed into electricity, the rest being wasted as heat. PV module efficiency decreases at a rate of approximately 0.40–0.65% with a one degree increment of module temperature. PV temperature can reach as high as 80 °C particularly in hot arid region. So, continuous efforts are being given to improving the efficiency of the photovoltaic. Photovoltaic thermal (PVT) system is a well engineered solar co-generation system amalgamating photovoltaic (PV) module and solar thermal components which provide both electricity and heat as end product. The PV cells suffer drops in efficiency with the rise in their operating temperature which may be improved by removal of the heat with proper cooling disposition. The waste heat thus removed to cool the PV module may be further utilized in heating applications like in crop and fish drying, textile industry, heating of buildings and regenerating dehumidify agent. The major advantage of hybrid PVT systems is that they offer a two fold yields, namely electricity and heat with a little extra cost and half of the space. This is an excellent example of optimization where the detrimental effect of overheating of the panel is being minimized by removing the extra heat and this waste heat is utilized for useful applications. There are various methods for cooling purpose such as micro-channel cooling system, thermo-electric cooling system, heat pipe cooling system, mist water cooling system, water film cooling system, PCM... (Sahay et al., 2013). All these systems work with different types of fluids and can be classified based in the next parameters: single phase system or two phase system, moving parts or not, with or without generation, active or passive, working fluid used... The commonly used methods of cooling systems are:

- Micro-channels: are the most typical confinement heat exchangers. This system has channels with a hydraulic diameter below 1 mm and especially applied in electronic cooling. They can be machined on the back of the sub-strates of microelectronic components in integrated circuits were the heat generated by the electronic components is transferred to the coolant by forced convection.
- Thermoelectric cooling systems: are based on thermoelectric devices which are composed of two semi-conductors: n type and p-type. The materials are attached thermally in parallel, but electrically in series. When a temperature gradient exists, the majority carriers diffuse from the hot side to the cold side setting up a voltage and the resultant current. In the reverse effect, an applied voltage forces a current through the materials causing an effective heat pump that cools one side and warms up the other. The hot side must be attached to a heat sink to dissipate the excess of heat (Jakhar et al., 2016).
- Heat pipes: The heat pipe transfer heat from a source through evaporation and condensation of the fluid (Brahim et al., 2014, 2013). Heat pipes are considered efficient heat transfer mechanisms which can transfer heat between long distances and its temperature drop is expected to be very. Several researchers collaborated with the application of heat pipes cooling technology in photovoltaic cooling. This method has particularly tried for concentrator photovoltaic panels (CPV).
- Phase change materials: Phase changing materials (PCM) are assigned to different ways, viz., based on their melting temperature or mode of their phase transition or their material composition, etc... The melting temperature of PCMs falls into two ranges: low melting temperatures that are below 200 °C and high melting temperatures which are above 200 °C. Investigations on the possibility of using phase change materials (PCM) as a medium of thermal regulation in photovoltaic (PV) systems is now a growing field of research all over the globe. PV-PCM system compared to a non-PCM system is positive everywhere in the world and in some regions it yields 6% excess energy on an annualized basis.

PV electrical efficiency is improved by cooling. This type of cooling can be classified in many ways depending on factors such as working fluid, number of phases, etc. A general classification of this type of cooling also called Flat plate PVT collector is showed in Fig. 1. Typical working fluid work could be air. The air is heated and it is used for space heating, water pre-heating, ventilation, or for industrial applications such as food drying. The majorities of designs are based in single pass or double pass that use the air as a working fluid in order to transfer the heat produced and decrease the operating temperature of the PV panels.

1.2. Hybrid water collectors

In Liquid PVT Collector, working fluid is water. Numerous studies of this cooling system have been developed in the last years. Works show that the efficiency increases notably by using water as the working fluid. The energy transferred by conduction toward the posterior surface of the panel is transferred to the water in the tubes. The configuration of the sheet and tube could have different shapes such as tubes, channels, and spiral tubes. Cooling systems have distinct limitations depending on the technology used for cooling. One of the common limitations is adhesive requirement because the PV module needs to be attached mechanically to the thermal absorber in order to have a better heat transfer. Another

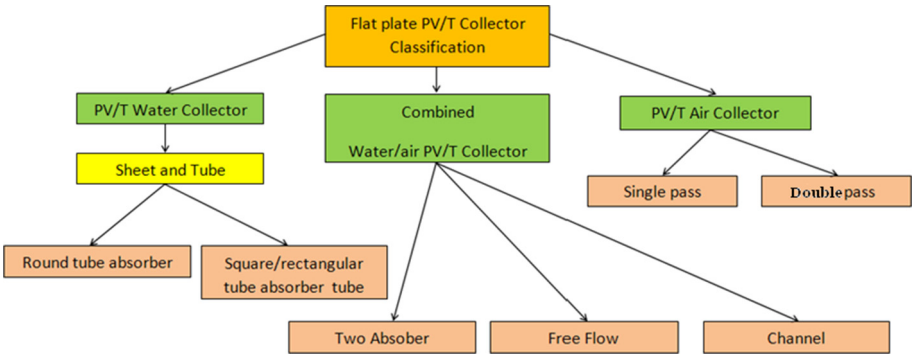


Fig. 1. PVT System classification (Ibrahim et al., 2011).

limitation is the stagnation temperature. The temperature of a common thermal collector can reach 200 °C, but the maximum temperature in a PV system could be 150 °C. For this reason, the typical stagnation temperature for the unglazed PVT is 80 °C, and for the glazed PVT it maybe 130 °C (Michael et al., 2015, 2016).

The first PV/T collector was considered by Wolf (Wolf, 1976) in 1976 combining solar PV and water heating system. The main concept was further improved in 1978 by Kern and Russell (Kern and Russell, 1978), studying its application in the climate of the United States. Hendrie (1979) presented an analytical model and experimental study for combined air-liquid PVT collector. Two years after, Raghuraman (1981) presented two separate one - dimensional models for predicting the thermal and electrical performance of PVT photovoltaic combined air/liquid PVT collectors, suggesting some recommendations to maximize the total energy gained from the collectors, Selective absorber Optical premasters, distance interval of glass absorber plate and thermal contact.

A new transparent hybrid a-Si PV/T was integrated and tested in the hybrid unit by Lalovic et al. (1986). The results obtained have shown the possibility of building simple and cheap hybrid systems with reliable photovoltaic and thermal efficiencies. The study on hybrid forced-air systems proved that the system can be autonomous only for specific design parameters and flows (Bhargava et al., 1991). This study was extended by adding a flat plane booster reflector to a flat hybrid solar air heater (Garg et al., 1991).

An important issue to achieve high thermal performance in a PVT collector is to make sure a good thermal contact between the photovoltaic element and the absorber plate. Usually, a layer of an adhesive with high conductivity is applied to bond the two parts. Different materials and application methodologies are evaluated to assess the thermal contact between the PV panel and the collector. Due to the poor thermal conductivity of the panel substrate material, a PV prototype panel was constructed using a metallic substrate with a thin insulating layer by Zakharchenko et al. (2004). It was noted that the power output of the PV panel increased 10% with the original design. More advanced techniques can be used, consisting on laminating together all the components (the transparent front glazing "not necessarily glass", the encapsulated material, the PV cells and the absorber). The experimental data also indicated a significant improvement for both thermal and electrical performance in comparison with other PVT concepts. Thermal and electrical efficiency for zero reduced temperature were 79% and 8.8%, respectively, using pc-Si PV cells (Zondag, 2008; Dupeyrat et al., 2011). A non-contact Photovoltaic-Thermal collector, which consists of a PV panel separated by a conventional sheet and tube solar thermal collector, was proposed by Khandelwal et al. (2007). At high values of PV transmissivity (> 0.75), the thermal efficiency of the non-contact type system exceeds that of the contact type collector at higher values of inlet temperatures.

Through a range of contact resistance from 0.005 to 0.05 °C m^2/W the effect of thermal contact was also studied by Siddiqui et al. (2012). From the case of ideal contact to the case of maximum contact resistance considered, the PV cell temperature increased around 18 °C. The absolute drop in efficiency was around 2% and about 19% of electrical power was lost due to the contact resistance.

Kalogirou (2001) reported that the average annual efficiency of photovoltaic power production increased from 2.8% to 7.7% with a hybrid system. The economic analysis included found a payback period of 4.6 years. Teo et al. (2012) found that electrical efficiency of a photovoltaic panel increased from 8.6% to 12.5%, when integrated with a solar air collector. The numerical model developed by Siddiqui and Zubair (2017) found that the use of hybrid collectors is more advantageous in climates with high solar radiation and ambient temperature, such as in the Middle-East region. Another advantage of PVT is their manufacturing cost. Thus, the energy payback time for PVT collectors was found to be considerably shorter than for discrete systems. Payback time shortened to 10 years on Greece for c-Si modules and 6 years for a-Si modules (Frankl et al., 2000; Tselepis and Tripanagnostopoulos, 2002). Concentrating sunlight is a key technique in order to reduce the system cost. With this solution, the area required for the PV modules decreased, and therefore payback time becomes shorter. However, PV module temperature increases with concentration which is disadvantageous for the electrical efficiency (Riffat and Cuce, 2011). It was ascertained that the minimum required cell area decreases with the use of boosters, and that high cost cells could be replaced by low cost reflectors. The improvement on the total efficiency of applying a booster diffuse reflector and glazing in a water hybrid system with pc-Si cells was registered by Tripanagnostopoulos et al. (1996) as well as the use of the glazing.

In order to assess the performance of a photovoltaic/thermal (PV/T) collector, the amount of electricity compared to the useful heat from the collector is an important factor. The performance of a PVT collector can also be evaluated in terms of exergy efficiency. PVT collectors are thermodynamically advantageous as they simultaneously generate high-grade (electrical) and low-grade (thermal) energy. Joshi and Tiwari (2007) concluded that there is an increase of about 2–3% in exergy due to thermal energy, in addition to 12% for the electrical exergy from a PVT system, which leads to an overall exegetic efficiency of about 14–15% for a PVT system. Sarhaddi et al. (2010) reviewed the literature for PVT exergy efficiency. It was found that the working fluid has a strong effect on the modified exergy efficiency, which can be increased if an incompressible fluid (water) is used in a PVT collector system.

In conclusion, use of PVT collectors can lead to both energy and economic benefits. However, research is still required, particularly on optimizing the PV cells layout, in terms of the fraction of the surface collector area covered by PV cells, and its preferable placement. An economic assessment of the PVT collector performance is essential to assert this technology option. PVT panels present an aesthetical and attractive solution, which can be applied into buildings, integrated on the facades and roofs. This technology is known as building integrated Photovoltaic/Thermal (BIPVT) systems, and has been studied by numerous researchers. This technology evolved from the building integrated photovoltaic (BIPV) technology Clarke et al., 1996; Sandberg and Moshfegh, 1998.

Typically, air systems are used, for convenience of installation and integrated with ventilation units, but also water collectors can be adapted, providing hot water and space heating simultaneously. Tripanagnostopoulos (2007) carried out an experiment study to increase the performance of a PVT system at University of Patras, Greece. A dual heat extraction operation system was investigated, through a combination of water and air PV/T. Three alternative models of passing the water inside the air channel were tested. To improving heat extraction using air, the modifications included placing thin metallic corrugated sheet into the middle of a flow channel, as well as attaching small ribs on the opposite wall of the PV module and installation of lightweight pipes along the channel. In order to improve the operation on a horizontal roof, a booster diffuse reflector was also combined with the PVT system.

Moradi et al. (2013) compiled and reviewed the major control parameters on the thermal/electrical performance of PVT collectors, such as packing factor, mass flow rate, PV solar cell materials, heat transfer fluids and geometries. Regarding the compiled information on the influence of the packing factor, it was found that a comprehensive knowledge of the variation of the packing factor and its effects with different fluids in different PVT systems still

does not exist. So, the study developed in his thesis helps to fill in the research on this aspect, with respect to water PVT collectors.

2. Component's characteristics of PVT flat plate water collectors

Most of the hybrid flat plate water collectors are sheet-and-tube type, originating from the most widely spread technology of thermal collectors. In this section, some major issues concerning the different components of a PVT collector are presented, which are correlated to the wide range of constructive designs of the hybrid collectors. Advantages and fragilities of diverse existing options will be addressed.

2.1. Transparent cover

The option for the number of transparent covers used in the sheet-and-tube configuration has been extensively reviewed by several authors (Fujisawa and Tani, 1997; Morita et al., 2000; Tripanagnostopoulos, 2010). It was noted that single glazed PVT collector presents higher thermal output than an unglazed one. Nevertheless, its electrical output is lowered reduced, due to additional optical losses (Fraisse et al., 2007). It was found also that the exegetics efficiency increases with the flow temperature up to a maximum value of about 13.36% for glazed and 11.92% for unglazed PVT collectors and the optimum flow temperature is, respectively, 83.6 °C and 38.8 °C (Morita et al., 2000). Chow et al. Chow (2009) carried out outdoor measurements on two similar sheet-and-tube thermosiphon PVT water collector systems in Hong Kong, in one glazed and another unglazed. The influence of six selected operating parameters was evaluated. The first law analysis indicated that the glazed design is always more suitable if either the thermal or the overall energy output is to be maximized. However, from the exergy analysis it was found that an unglazed system is more advantageous for high values of PV cell efficiency, packing factor, water mass to collector area ratio, and wind velocity, whereas the increase of solar radiation and ambient temperature are favorable for a glazed system. The use of more covers creates additional losses by reflection. Instead of glass, other lighter, cheaper and stronger materials, such as polycarbonate, polymethyl-methacrylate, polyvinyl fluoride, can be adopted for the cover. However, the use of glass is the best option, because of its good optical properties, resistance to UV and high temperatures. The air gap between the PV laminate and the cover material must be thin enough to benefit from the insulating properties of air, and preventing at the same time convective flows and micro turbulence. Generally the gap width should be between 15 mm and 40 mm (Gordon and J., 2001; Aste et al., 2014).

2.2. PV modules

Photovoltaic modules are applied to an absorber typically use EVA for the encapsulation of the cells. However, there are certain technical difficulties related to standard EVA lamination of PV modules. It can decompose in the presence of acetic acid (delaminating, coloration, degradation of PV cells by acid) at temperatures above 80 °C. Thus, conventional PV laminates cannot withstand stagnation temperatures in glazed collectors usually operating between 120 °C and 180 °C. A novel glazed PVT collector concept based on PV laminate with siloxane gel is now under development at the Czech Technical University in Prague. Application of siloxane gel instead of EVA lamination compound offers several important advantages, such as high temperature resistance, high transparency, compensation of thermal dilatation stresses and favorable heat transfer from PV to heat exchanger in PVT collector (Matuska, 2014; Poulek et al., 2012).

2.3. Absorber plate structures

In a PVT solar collector, the absorber plate is an essential functional element, since it transfers the solar energy that was not converted to power by the photovoltaic element, to the fluid. Its desirable properties are higher thermal conductivity and lower specific heat capacity. For these reasons, absorber plates for water PVT are generally metallic, such as copper, aluminum or more rarely steel. The upper side of the plate should allow the perfect adhesion of the cells or the PV laminate, thus enhancing the heat removal from the photovoltaic component. Besides the most usual sheet and tube arrangement, other structures can be found for absorbers in water hybrid FPC, as showed in Fig. 2.

The designation of "roll bond" comes from its production process. In this manufacturing technique, a sandwich of two aluminum sheets is formed by means of a special hot or cold rolling process. Before pressing together the aluminum sheets, the desired pattern of channels is printed with a serigraphy process on the inner surface of one sheet. A special ink is applied to prevent subsequent welding of the surfaces along the pattern. This allows flexibility to configure the desired channels profile, according to various configurations, while maintaining a low production cost (Aste et al., 2014; Bombarda et al., 2016).

The "box channel" structure is formed by parallel ducts with a rectangular cross section. It can be made from an extruded or pultruded profile, that is generally aluminum, or, occasionally, using polymeric materials, such as rubber or fiberglass (Cristofari, 2012; Cristofari et al., 2002). However, the relatively large coefficient of thermal expansion of plastics hinders the binding with the PV laminate, and their low thermal conductivity and temperatures resistance make them limited to this application. Special components are required to connect the inlet and the outlet manifolds to the rectangular channel, which greatly increases the costs and the technical issues. The manufacturing process of the box channel configuration is relatively expensive (Van Helden et al., 2004).

A strategy to enhance the deployment of hybrid collectors is by reducing its production cost, without prejudice of the performance. A numerical model developed and validated by Charalambous et al. (2011) optimized the absorber plate configuration, by reducing the size of the tubes and the amount of metal worn in the flat panel construction, for both harp and serpentine configuration. Based on their results, the cost of expensive raw materials such as copper can be reduced.

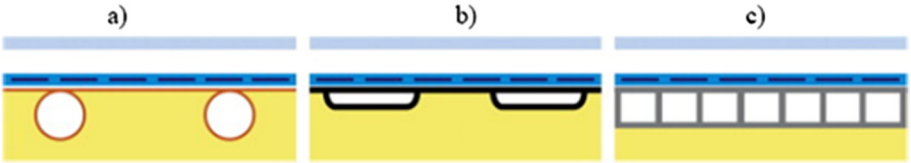


Fig. 2. thermal absorber structures: (a) sheet and tube, (b) roll bond and (c) box channel Aste et al., 2014.

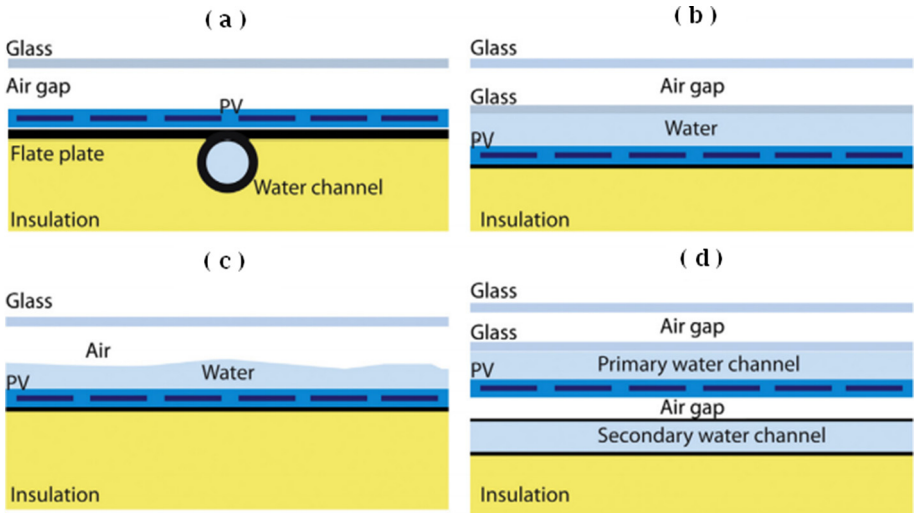


Fig. 3. Different configurations for flat plate water hybrid collectors (a) sheet and tube; (b) Channel; (c) Free flow; (d) Double absorption (Charalambous et al., 2007).

2.4. Configurations of the collector structure

Beyond the traditional "sheet and tube" type of water hybrid collectors, some other designs have been evaluated (Matrawy and Farkas, 1997; Fan et al., 2005, 2007; Fan and Furbo, 2008). The performance was investigated by varying the layout of the different parts (PV module, absorber, fluid flow). Four main types of PVT were identified according to the water flow pattern and the heat exchange method, as illustrated in Fig. 2: sheet and tube, channel, free flow and double absorption.

It is obvious that a more efficient heat transfer is obtained when the mean distance between heat generation and heat collection is minimal (Aste et al., 2016). This is realized when the liquid flows directly over, or below, the PV cells, in the channel concept. Fig. 3 shows channel flow over the PV cells. In channel PVT collectors, the absorption spectrum of the fluid should be sufficiently different from the absorption spectrum of the PV. The existence of an additional glass cover turns the assembly heavy, and fragile.

The PV module can be opaque, or transparent with a separate black thermal absorber underneath the channel. This geometry is better suited to withstand water pressures in the channels than in the case of a broad channel. The absorber "box channel" structure is applied in this configuration. Sandnes and Rekstad (2002) have used this type of design, filling the square-shape box-type absorber channels with ceramic granulates, in order to improve the heat transfer to flowing water. The results for applications to low-temperature water heating systems were promising.

The channel design with liquid flow under the PV cells has been used in the BIPVT technology, for facades (Ji, 2006). With the use of wall-mounted water-type PVT collectors, the system not only generates electricity and hot water simultaneously, but also improves the thermal insulation of the building envelope. The simulation results stated that there is an optimum water mass flow rate for the desired energy performance. A dynamic simulation model of a building-integrated photovoltaic and water-heating system, was developed by Chow et al. (2006), having the same absorber design. Their study was further they analyzing the annual energy performance of a BIPVT water system, in both natural and forced circulation modes. Compared to a normal building facade, both modes of operation were in a position to reduce the thermal transmission through the PVT water wall by about 72% and 71%, respectively. An economic evaluation was also performed by determining the payback period, and the economic advantage of the BIPVT water system was proved to be much better than a simple BIPV. Cristofari et al. (2009) also developed a simulation model for a channeled type water PVT collector manufactured in a copolymer material. The system was suitable for wall integrated water-heating.

Recently, the use of flat-box absorber design on PVT systems has become more frequent (He et al., 2006). A model for flat-box aluminum-alloy photovoltaic and water-heating system designed for natural circulation was developed by Ji et al. (2007). The simulated results indicated that the higher the packing factor and the glazing transmissivity, the better the overall system performance.

In a free flow design (Fig. 3c), the cooling effect of the thermal fluid moves toward the front surface of the PV panel. One glass layer is eliminated, compared to the top channel flow design, whereby reflection losses and material costs are reduced, as well as the mechanical problem of breaking the glass cover is avoided. A disadvantage of this design is the increased heat loss due to evaporation. As in the case of the channel flow type, the fluid flowing over the PV panel needs to be transparent for the solar spectrum (Zondag et al., 2003). Until today, this configuration had only been studied on a theoretical basis.

Zondag et al. (2003) evaluated nine design concepts of hybrid water collectors, based on the four types. Numerical results revealed better thermal efficiency for the two absorber design (Fig. 3d). Regarding the total annual yield for a domestic hot water system, the best option was the channel-below-transparent- PV design. The one-cover sheet-and-tube design was also proved to be a useful alternative, since its efficiency was only 2% inferior. As this latter design accounts for the simplest to manufacture, the single cover sheet-and-tube design seems the most promising of the examined concepts for domestic hot water production.

The two-absorber panel in Fig. 3d combines, to a certain extent, the channel and the free flow solutions. A transparent PV laminate is used as a primary absorber below the primary water channel. A black metal plate is installed as a secondary absorber, separating air from the secondary water channel. The water passing through the upper channel is returned through the lower channel. This design reported a high thermal efficiency (Zondag et al., 2003). A main benefit of this concept is that a lower mean PV cell temperature is maintained, compared with geometries with heat and electricity generation in one plane (Fig. 3). A disadvantage, however, is the complexity of the geometry that makes the module difficult to manufacture.

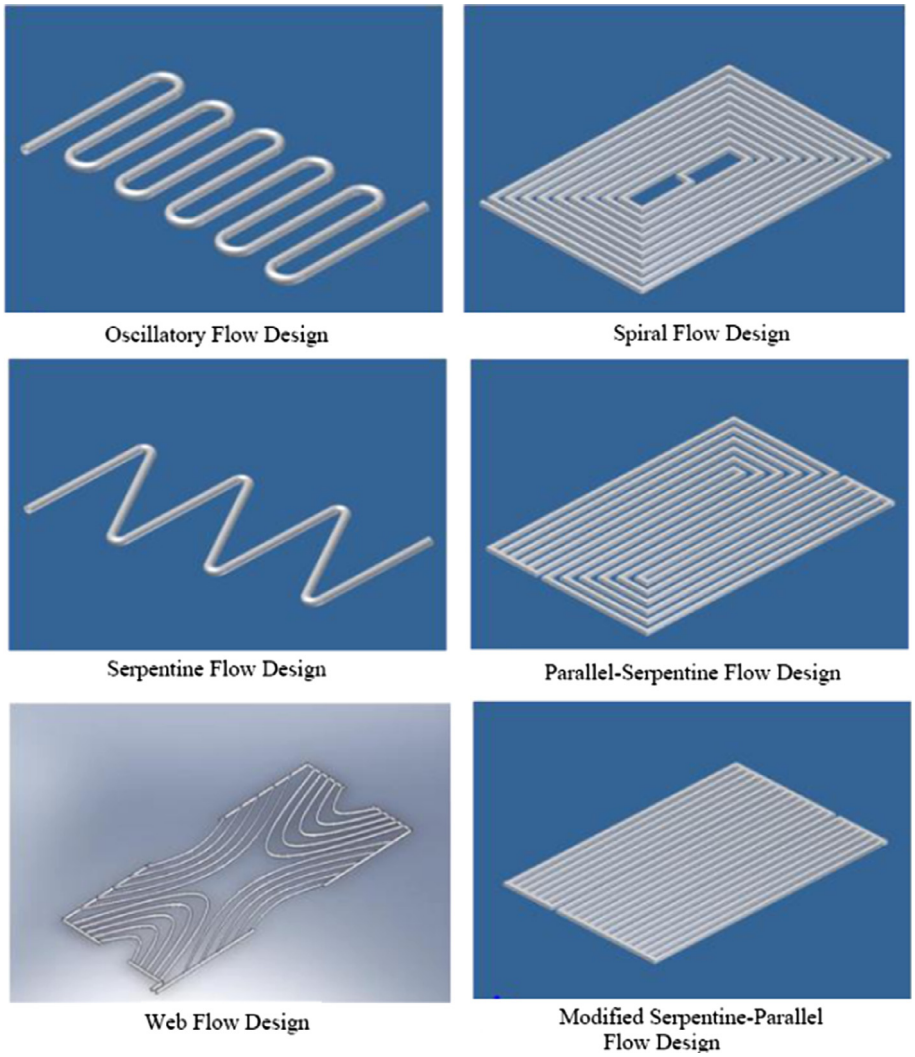


Fig. 4. Different types of paths for the flow in PVT collector.

Seven configurations of serpentine collectors with single glazing were designed, investigated and compared through simulation methods by Ibrahim et al. (2011). Different types of paths for the flow were considered: direct, oscillatory, serpentine, web flow, spiral, parallel-serpentine, and modified serpentine-parallel. The best results for the efficiency were found in the spiral flow design shown in Fig. 3.

The pressure drop across the flow is another important factor, because of mechanical and thermal aspects. A larger pressure drop requires higher energy demand on the circulation pump. Recently, experimental result carried out by Pieper and Klein (2011) and Hermann (2011) with a configuration called bionical, were compared to the harp and serpentine configurations. The plate used in the bionical configuration replicates the structure and morphology of the blood vessels: small size parallels channels connected together in pairs into a bigger channel and then they are all collected in the manifold, as showed in Fig. 4. The results showed that the harp configuration gives a better response than the serpentine arrangement, and slightly worse than the bionical, for high flow rate (Kremers and Pieper, 2015).

In overall, a water-based PV/T performance is largely dependent upon water temperature, flow rate, water flow channel geometry and sizes, PV type, as well as external climatic condition. Researches related to water-based PV/T systems usually focus on:

(i) Determining appropriate water flow velocity and temperature,
(ii) Optimizing water flow channel's geometry and sizes,
(iii) Suggesting the configuration of the integrated PV/T panels including covers, PV cells and their connections, etc.

Some technical difficulties are presented to improve water-based PV/T systems like the water temperature variation over the operating period and the freezing problem at a cold climate region.

3. Parameters affecting the performance of PVT flat plate water collectors

The development of theoretical and empirical models is a key tool in engineering to predict the performance, such as from solar PVT collectors, over a wide range of operating conditions. After validation, these models can be utilized, e.g. for design purposes. Analytical models of PVT collectors depend on energy balance equations, considering the relevant heat transfer modes and adequate boundary conditions. Several software tools are available for solving the governing equations, such as EES, or ANSYS Fluent (ANSYS). EES is a general equation-solving program that simultaneously solves a set of non-linear algebraic equations. ANSYS Fluent software uses finite volume method discretization of a set of

Table 1
Thermal and electrical efficiencies from selected works on water PVT collectors (Aste et al., 2014).

Plate type	PVT type	Flow rate (kg/s m^2)	Thermal efficiency	Electrical efficiency	Analysis type	Refs
Sheet and tube	Uc	0.02	66%	14%	Exp	Kim and Kim (2012a) and Zondag et al. (2003)
	C	0.02	58%	8.9%	Num	
	Uc	0.02	52%	9.7%	Num	
Box channels	Uc	0.02	70%	15%	Exp	Chow et al. (2006), Ji et al. (2007), and Sandnes and Rekstad (2002)
	C	0.02	57%	12%	Num	
	C	N/A	45% daily	10.15%daily	Exp	
	C	0.02	60%	9%	Num	
	C	N/A	71%	N/A	Exp	
	Uc	N/A	76%	N/A	Exp	
Roll bond	C	0.01	49.3%yearly	10.3%yearly	Num	Bai et al., 2012; Dupeyrat et al., 2011
	C	0.02	79%	8.7%daily	Exp	

C: Covered; Uc: Uncovered; Exp: Experimental; Num: Numerical.

differential equations to m, typically describing fluid flow and heat transfer problems. TRNSYS (Transient System Simulation Tool) (TRNSYS) is used for unsteady analysis, simulating the performance of a modeled system over a period of time. It integrates information about the variation of several environmental conditions, such as radiation, wind and ambient temperature, for a certain geographic location.

PVT efficiencies depend on several parameters; a summary of published values of electrical and thermal efficiencies of water PVT collectors is listed on Table 1, according to studies selected by Aste et al. Aste et al. (2014). Table 1 was outlined evidencing the influence of the flow rate by surface area of the absorber, absorber plate configurations and existence of cover. The reported efficiency values correspond to zero reduced temperature, except when otherwise specified.

3.1. Packing factor (P_f)

The application of PV cells over an absorber plate obstructs the incident radiation, causing a reduction on the energy conversion to useful heat, as confirmed in several studies. Wu et al. Wu et al. (2011) developed a model to study the performance of a heat pipe PVT system, and used it for carrying out a parametric investigation on numerous parameters, including P_f. It was noted that the solar cell temperature decreased, by increasing P_f from 0.7 to 0.9. Results obtained by Dupeyrat et al. (2011) confirm that a higher packing factor has a positive impact on the PV efficiency, with an increase of about 3.5%, but impairs the thermal efficiency. Garg et al. Garg and Agarwal (1995) analyzed the performance of a direct forced flow hybrid system for domestic hot water production, using a simulation model. The total thermal energy was extracted from the useful heat stored in a tank. They concluded that increasing the packing factor from 50% to 100%, the total daily efficiency increased. The efficiency was defined as the ratio between the absolute total thermal plus electrical energy over the total solar insolation. Dubey and Tiwari (2008) developed a thermal model of a PVT solar water heating system and applied it for different values of P_f = 30.56%, 50% and 100%. The PV modules were applied at the water inlet side of the collector, where they substituted the glass cover, since PV cells were encapsulated in glass. A decrease in thermal efficiency was observed with the increase in the area covered by PV cells (Dubey et al., 2009). The results were integrated and compared by Tiwari et al. (2011) through analysis of the hourly variation of the cell temperature and solar cell efficiency during one day. Chow et al. Chow et al. (2006) used an experimentally validated numerical model to compare thermal and electrical efficiencies of a photovoltaic-thermosiphon collector system, for different packing factors of 0.50% and 100%. A box channel structure was built from a multiple of extruded aluminum alloy modules for the absorber. The solar cells were encapsulated between 2 layers of EVA and tedlar-polyester-tedlar. The position of the PV module was also evaluated. The results for thermal and electrical efficiencies were better with the application of the PV module in the lower part of the collector than in the superior one.

The results of Herrando et al. (2014) with low solar irradiance and low ambient temperatures in the UK show that a complete coverage of the solar collector with PV, and a lower collector flow-rate, benefit both heat and electrical conversion achieved with the PVT collector, while maximizing the CO_2 emissions savings. It was found that with a fully covered collector and a flow rate of 20 l/h, 51% of the total electricity demand and 36% of the total hot water demand over a year can be supplied by a hybrid PVT system. The electricity demands coverage value was only slightly higher than an equivalent PV-only system (49%).

Despite studies on the influence of the packing factor in the thermal and electrical already exist; references on the effect on the overall efficiency are scarce (Chow et al., 2006), and more particularly the specific influence on the loss coefficient of hybrid collectors.

3.2. Geometric characteristics, environmental and operating conditions

Bergene and Lovvik (1995) defined a physical model for a hybrid flat plate collector with finned water channel. A parametric study was carried out to study the impact of factors such as fin width to tube diameter ratio, inlet temperature and mass flow-rate, on the thermal and electrical efficiency. They verified that the thermal efficiency is approximately halved when the fin width to tube diameter ratio is increased from 1 to 10. The flow-rate and the inlet fluid temperature were identified as the most important

Table 2
Optimum flow rates selected for PVT types.

Channel type	PVT type	Flow rate (kg/s m^2)	References
Parallel	Covered	0.0027	Nualboonrueng et al., 2013
Parallel	Covered	0.0014	Kalogirou, 2001
Roll bond	Covered	0.0014–0.0049	Morita et al., 2000
Parallel	Covered	0.005	Chow, 2003
Parallel (square)	Covered	0.0054–0.0064	Gao, 2010
Parallel	Covered	0.015	Garg and Agarwal, 1995
Box	Covered BiPVT	0.025–0.04	Ji, 2006

parameters affecting electrical efficiency, while the total efficiency was strongly dependent on the fin size.

The existence of an optimal mass flow-rate for maximum thermal, electrical and total efficiency was identified in the already referred study by Garg et al. Garg and Agarwal (1995). The effect of the capacity of a water storage tank was further studied. It was concluded that a smaller volume improved the thermal efficiency only, but the electrical and total daily efficiency reduced. Other authors also mentioned the existence of an optimum flow-rate (Kalogirou, 2001; Morita et al., 2000; Garg and Agarwal, 1995; Garg and Adhikari, 1997). Daghigh (2011) studied solar PVT modules integrated into the building structure (BIPV/T). The effect of mass flow-rate on the system efficiencies was evaluated, for different types of cells. Although an optimum value was not verified, it was found that efficiencies increased up to a given value of flow-rate, and then stabilized. The effect of the collector flow-rate on the performance of PVT systems was also studied by Herrando et al. Herrando et al. (2014), It was observed that the electrical performance did not change notably (<5% variation) with the collector flow-rate, whereas the hot water production was significantly affected. It dropped by about 35% as volumetric flow-rate increased from 20 l/h to 200 l/h. Table 2 shows reviewed published optimum flow rate values reported in the literature, for different absorber plate structures and PVT types (Aste et al., 2014).

In order to conduct an inquiry into parameters for optimizing PVT hybrid collector, exergy performance using second law approach was used. Besides the presence or not of the cover glass, the mass flow rate was found to a key parameter affecting the exergy efficiency. Respecting thermal and global exegetics efficiencies, there is a relatively low value of the optimum flow rate of 2.3 g/s m^2 for glazed PVT collectors. The photovoltaic conversion efficiency under a wide range of environmental conditions decreased with the solar radiation intensity (3.6%/kW m^{-2} at T_a = 20 °C); however, it was observed that the exergy efficiency of PVT increased 3.6% per kW m^{-2} (at T_a = 20 °C). Therefore, the advantage of PVT will be more remarkable for the solar radiation (Fujisawa and Tani, 1997).

As solar energy is intermittent, many algorithms and electronics were designed to identify the maximum power generation from PVT collectors. Nevertheless, no control systems have been introduced in order to track maximum power generation from PVT system. Ammar et al. (2013), suggests a PVT control algorithm based on Artificial Neural Network (ANN) to adjust the Maximum Power Operating Point (MPP) by considering PVT model behavior. An optimum mass flow rate is calculated for a considered irradiation and ambient temperature. Simulation results demonstrate remarkable concordance with ANN outputs.

Another parameter that affects thermal and electrical performance of hybrid collectors is the operating water temperature. The electricity production from PV cells is favored by low temperatures, whereas the value of the thermal energy is higher at higher temperatures. Evola and Marletta (2014) demonstrated that, for any operating condition of solar irradiance and flow rate, it is possible to calculate an optimum water inlet temperature that maximizes the total exergy generated by the system. The optimum temperature falls within the range commonly occurring in solar thermal systems, and can be obtained in practice through a simple feedback control system.

The effect of water inlet velocity and inlet temperature on the performance of a PVT collector was analyzed by Siddiqui et al. (2012). Fig. 5 shows the results of thermal and electrical performance for the variation of the inlet velocity in a range from 0.01 m/s to 0.1 m/s. For this range, the average PV cell temperature decreased from 41.1 °C to 30.6 °C and the water outlet temperature dropped from 30.7 °C to 25.7 °C (Fig. 5a). The increase in electrical efficiency for this range was of about 1.2%, with a minimum value of 9.4%. The results for the influence of inlet temperature of the fluid, ranging from 4 to 45 °C, are shown in Fig. 6. Average PV cell temperatures increased from 14.5 °C to 50.1 °C (Fig. 6a) and the electrical efficiency dropped from 12.28% to 8.4% (Fig. 6b). It would be interesting to complete this study with results for the global efficiency, to understand the relative weight of each parameter.

3.3. Methods to optimize PVT performance

A genetic algorithm has been used by Sobhnamayan et al. (2014), to optimize the exergy efficiency of PVT water collector, for a selected group of environmental and design conditions. Effects of water inlet velocity at the pipe diameter, solar radiation intensity and wind velocity were examined. The exergy efficiency has extreme points and showed points of global maximum, for a pre-selected range of operational and design conditions. The optimum conditions were identified by the maximum values of exegetic efficiency for each parameter, with values for inlet water velocity and pipe diameter of 0.09 m/s and 4.8 mm, respectively. Maximum exergy efficiency was found to be 11.36%.

Recently, mathematical techniques were to be incorporated into algorithms for *Multi-Objective Optimization (MOO)*, in order to overcome the generic problem of global optimization. Vera et al. (2014), studied simultaneously the design and performance of water cooled PVT systems, utilizing an elitist multi-objective evolutionary algorithm non-dominated sorting genetic algorithm-II (NSGA-II). NSGA-II derives a Pareto optimal set, which illustrates the trade-off between solutions. The study focused in the following selected parameters: water mass flow rate, length of the collector, packing factor and air gap thickness. PVT glazed and unglazed configurations were modeled. Electrical and thermal efficiencies are the two objectives functions to be maximized. The mathematical analysis was performed based on results for a 24^h simulation obtained using a TRNSYS dynamic simulation tool. The dynamics of the results are complex, but prove that this approach can provide a good tool to understand the relative weight of each parameter in optimizing a scenario.

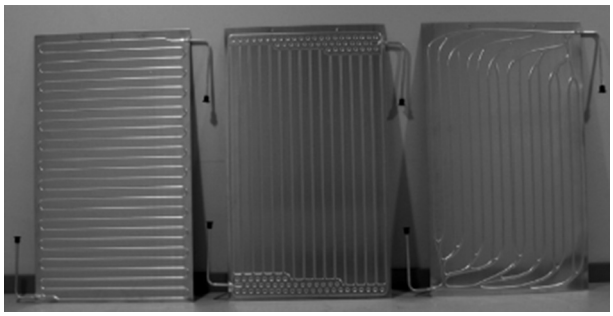


Fig. 5. Serpentine, harp and Bionical configurations.

4. Hybrid air collectors

PVT modules working with air are utilized when there is a demand on hot air for applications like drying and preservation of agricultural crops, dehydration of industrial products and space heating, increased ventilation, as well as electricity generation. Some advantages over water collectors are the reduced corrosion and leakage, no freezing, no need for high pressure protection, lighter weight and easiness of installation (Kumar and Rosen, 2011). However, the thermal output is inferior to the water collectors because of poor heat transfer between the absorber plate and the flowing air, and lower density per unit mass. Flat plate air

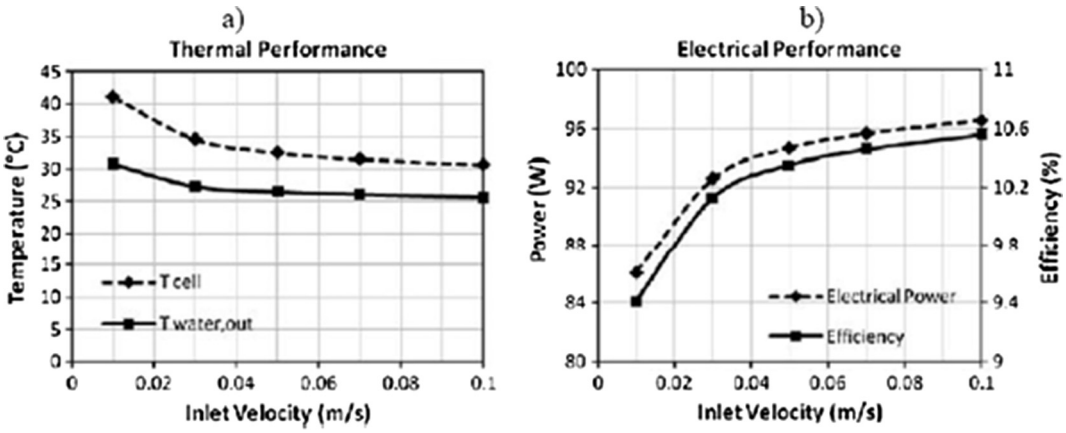


Fig. 6. PVT Collector performance variation with heat exchanger fluid inlet temperature (Siddiqui et al., 2012).

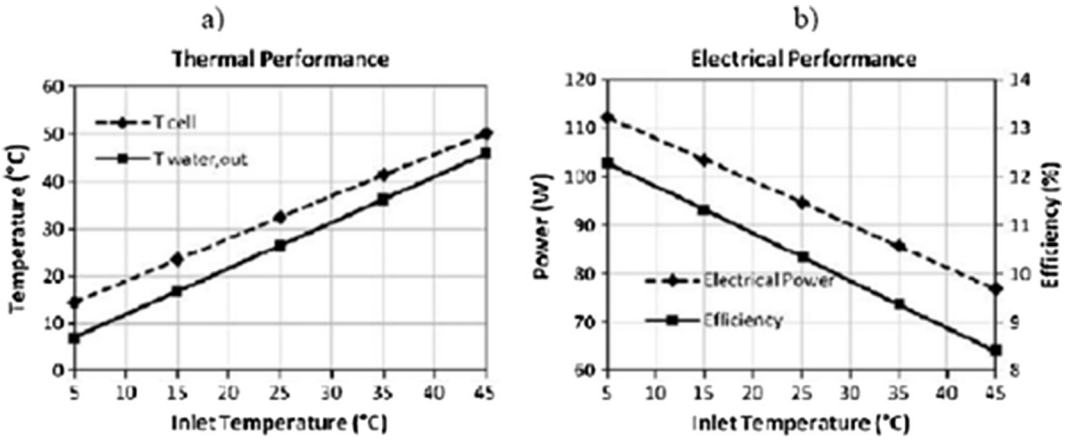


Fig. 7. PVT collector performance variation with heat exchanger fluid inlet temperature (Siddiqui et al., 2012).

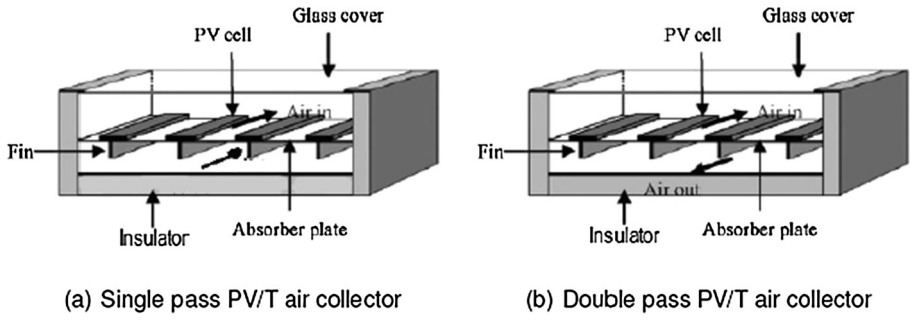


Fig. 8. Single and double pass PVT air collector (Ibrahim et al., 2011).

collectors exist in different designs, which the most common models are shown in Fig. 7. The majorities of designs are based in single pass or double pass that use the air as a working fluid in order to transfer the heat produced and decrease the operating temperature of the PV panels. They basically differ in terms of the position of the cells relative to the air channel and the number of streams. Photovoltaic cells are placed over the black absorber plate that should not be selective (Hegazy, 2000).

Bhargava et al. (1991) proposed a single-pass PVT air heater, with air flux under the absorber, similar to the represented in Fig. 8b. Sopian et al. (1996) compared its performance to a new collector, using steady-state models with a double passage design, as showed in Fig. 8d. Performance analysis showed that the double-pass photovoltaic thermal solar collector produced better performance over the single-passage design within normal collector mass flow rate range 100–300 kg/h. However, the increased in energy demand for circulating the air through double-pass model was not considered. In their later work, a mathematical model for the evaluation of the performance of the double collector was developed and experimentally validated (Sopian et al., 2000).

Hegazy (1999) defined a simple criterion for maximizing the useful heat based on the channel geometry, for thermal collectors with single pass configuration, with flow under the absorber. An optimal value for channel depth-to-length ratio of 0.025 was determined. In his extended work (Hegazy, 2000), he compared the thermal, electrical, hydraulic and overall performances of the

designs presented in Fig. 7. The effects of air flow rate per collector surface unit area and selectivity of the absorber plate and PV cells on the performances were analyzed, based on the developed models. The main conclusions are:

- For a particular design, the thermal efficiency is enhanced with the increase of the air specific mass flow rate. The available net electrical energy, resultant from the deduction of the power needed for circulating the air, from which, available net electrical energy, significantly decreases with air specific flow rate;
- Design corresponding to Fig. 7a presents the lowest overall performance, while the other configurations exhibit comparable performance up to a specific mass rate of 0.02 kg/s m^2. For higher values of m/A, design (c) has the highest overall performance, followed by the layout (d). For each design, there is an optimal mass flow rate beyond which overall performance of the collector decreases;
- Performance comparisons indicate that the design (c) is the most suitable for converting solar energy into low grade heat and high quality electrical energy, while also it is of simple construction.

Tonui and Tripanagnostopoulos (2007a, 2007b, 2008) worked on the improvement of the heat transfer rates for air hybrid collectors by the enhancement of the air channel layout (Fig. 8). In Fig. 8a, a thin flat metal sheet is suspended in the middle of the air channel. Within Fig. 8b, fins with rectangular profiles are included with the back wall of the air duct parallel to the flow direction. Glazed and unglazed versions were analyzed. Both experimental and theoretical results showed that the suggested modifications improve the operation of the PVT air system. Parametric studies were performed to analyze the influence of channel depth, channel length, air flow rate and temperature, incident radiation, collector tilt angle and exit vent on the thermal and electrical performance of the hybrid collector. The parametric analysis showed that the induced mass flow rate and hence thermal efficiency decrease with increasing inlet temperature and increases with tilt angle for a given insulation level. The results also showed that there is an optimum channel depth for which the mass flow rate and thermal efficiency are in maximum. For the studied systems, the optimum channel depth occurred between 0.05 and 0.1 m, with both modified systems showing slightly higher optimum depth. The thermal performance also increased with an exit area of the channel, which should not be restricted and made as large as possible, or equal to the duct cross-sectional area.

The effect of the fins, depth of the ducts, flow rate, inlet temperature and packing factor in thermal and electrical efficiencies was also evaluated by Kumar and Rosen (2011), for a double- pass configuration, but with the fins arranged perpendicular to the direction of air flow to enhance the heat transfer rate and efficiency. The extended fin area reduced the cell temperature from 82 °C to 66 °C. The advantage of a high packing factor was also proved, as it led to a better electrical output per unit collector area, and better control of the cell temperature, despite marginally reducing thermal output.

Tiwari and Sodha (2007), studied different layouts for the single pass configuration with air flow under the absorber. Single glazing and the use of tedlar on the back of the solar cells were also considered. It was observed that a glazed hybrid PVT without tedlar gave the best performance. Dubey and Tiwari (2008) and Dubey et al. (2009) developed in their works analytical models to evaluate electrical efficiency of air PVT collectors, as a function of climatic and design parameters. They compared the difference in the behavior of glass to glass modules with glass to tedlar modules with no glazing. They also investigated the effect of an inferior air flow and found that the best electrical efficiency was achieved with glass to glass modules with channels.

Garg and Adhikari (1999) investigated an air PVT collector coupled with a compound parabolic concentrator (CPC). They observed that concentrating PVT collectors provide higher efficiency compared with the systems without a CPC. A parametric analysis of the thermal and electrical performances of the system showed that the thermal and electric output increased with the collector length, air mass flow rate and packing fraction, and decreased with the channel depth. It was also observed that the system coupled with CPC always perform better in terms of both the thermal and electric output. The combination of alternative techniques that enhance system performance is a useful strategy to find improved solutions. Othman et al. (2005) studied the inclusion of CPC through and fins in a double pass configuration of a PV/T-air collector with a layout presented in Fig. 9a. The fins are attached to posterior side of the absorber plate. It was pointed out that the electrical performance was significantly influenced by the air flow rate and temperature. The work was extended to a similar configuration, without the CPC through (Othman et al., 2007) as showed in Fig. 9b). The height of the upper channel was fixed, but the height of the channel below the absorber plate could be set to observe its effect on the system performance. The experiments indicated that using fins as the integral part of the PV module increased the overall efficiency.

An unglazed, single pass configuration with air flow under the absorber was also tested by Othman et al. (2009). This unit had an aluminum ∇-grooved absorber plate in the air channel, as depicted in Fig. 10. Results of the experiment showed that, although the electrical efficiency had been only improved by 1%, the thermal efficiency enhancement was high by 30%.

The use of a porous media was applied by Sopian et al. (2009) as showed in Fig. 11 to a double pass prototype at the lower channel. Experimental data proved that this technique led to the increase of the thermal efficiency of the system as high as 60–70%. Such design was concluded to be suitable for drying applications.

Exergy analysis of a hybrid solar collector for air heating was conducted by Sarhaddi et al. (2010). The thermal, electrical and overall energy efficiency was found to be about 17.18%, 10.01%, 45%, respectively. The exergy efficiency was 45% under a simple climatic, operating and design parameters. Optimized values of air inlet velocity and irradiance were identified. It was noted that the modified exergy efficiency depended slightly on air inlet temperature or duct length, while increasing wind speed led to an increase in the exergy efficiency.

The use of air for building integrated hybrid collectors (BIPV/T) is more usual than water. Sandberg and Moshfegh (2002) studied characteristics of the air gaps behind solar cells located on vertical facades. Mass flow rate, velocity, temperature rise and location of the air gap were the parameters being verified and measured. Muresan et al. (2006) also developed detailed models for describing the heat transfer modes in the air channel of PV module integrated in facades. A parametric study was done for a case of a vertical channel heated at one side, as a function of the channel width, wall heat flux, and dimensionless turbulent intensity. PVT hybrid exterior clapboard-shaped wallboards have been tested experimentally by Nagano et al. (2003) permitting modular assembly, which simplifies the installation. A scheme of the system is illustrated in Fig. 12. The air gap created between the hybrid wallboard and the thermal insulation of the exterior wall can be used for solar heating the ventilation air as well as the tap water feed of a hot water system. The study presented an analytical and experimental evaluation of both the electrical power generating ability and the solar heat collection capacity of six variations of the experimental PVT hybrid wallboard during winter. In addition, exergy analyses were conducted.

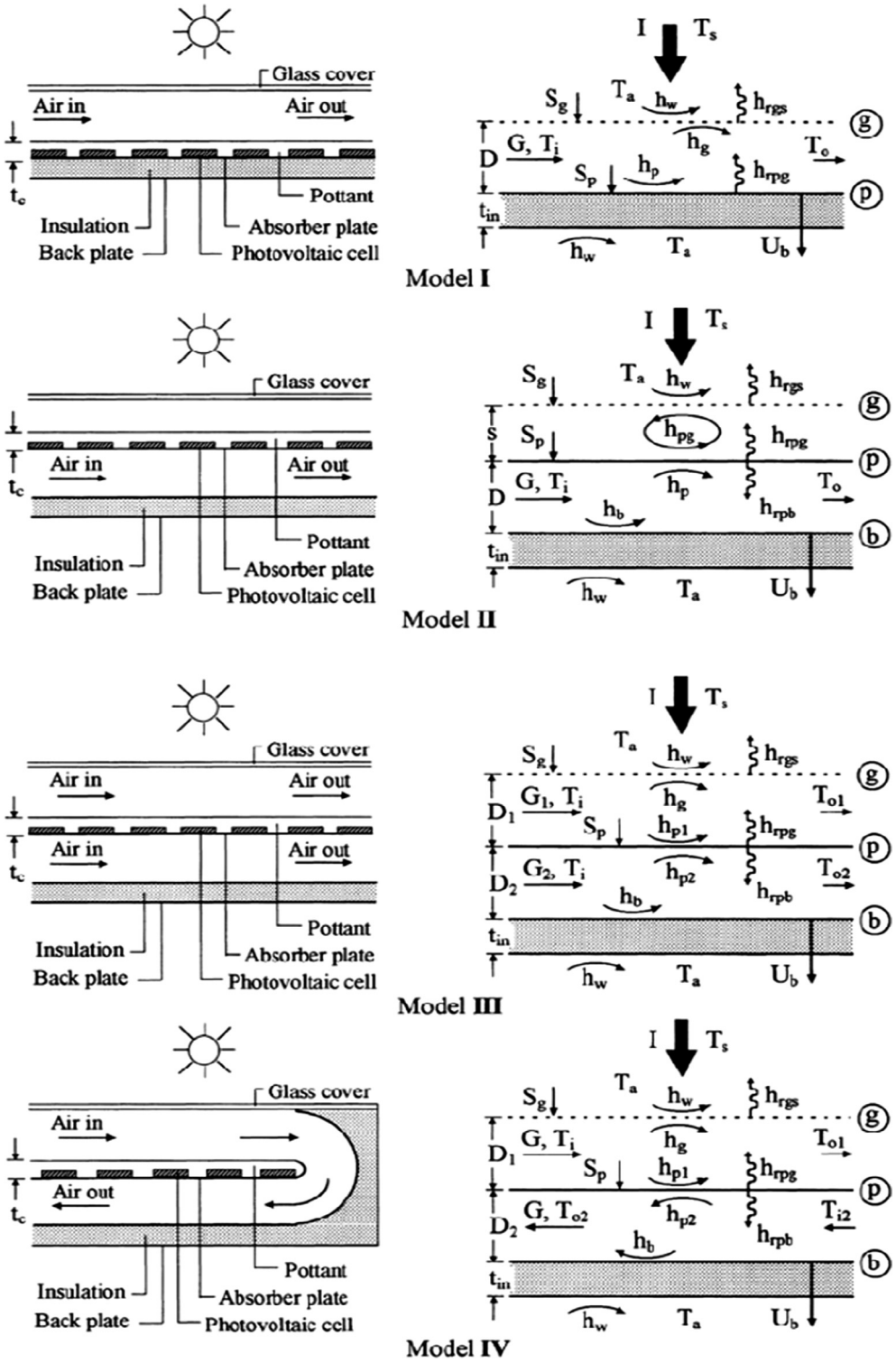


Fig. 9. Schematics of the various PVT models: (a) single pass, air flow over absorber; (b) single pass, air flow under absorber; (c) single pass, both sides of absorber; (d) double pass Hegazy, 2000.

Recent R&D in PV technology improved transparent PV cells. The use of this type of cells is particular interesting in BIPVT applications. Research study carried out by Guiavarch and Peuportier (2006) pointed out that, by using semi-transparent pc-PV modules as a cover for an air collector, the ventilation air can be pre-heated and the global efficiency can reach 20%, 6% higher than the reference case. Vats et al. (2012) studied the effect of packing factor of semi-transparent PV modules integrated on the roof of a building. Different PV technologies for the PV module were analyzed, in energetic and exergetic basis analysis: mono-crystalline silicon (c-Si), poly crystalline silicon (pc-Si), amorphous silicon (a-Si), cadmium telluride (CdTe), copper indium gallium diselenide (CIGS), and a heterojunction with thin layer (HIT). Maximum annual electrical and thermal energy conversion was measured for a-Si PV

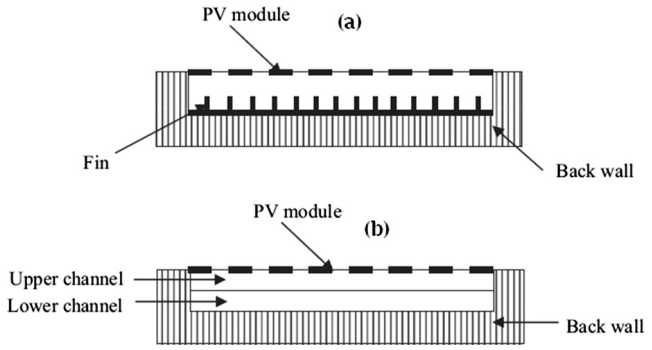


Fig. 10. Configuration improvements to single pass inferior flux air PVT collectors (Kumar and Rosen, 2011).

module with a packing factor of 0.62. It was observed that a decrease in the packing factor from 0.83 to 0.42 led to a decrease of the module temperature by 10 °C and increase of its efficiency by 0.2–0.6%. A PV module with 0.62 packing factor was found to perform better than the one with 0.83 packing factor, for both electrical and thermal conversion.

Currently, there are commercially available unglazed air types of PVT collectors. However, the application of air heating in the residential market is restricted. A market study was carried out by Hussain (2013) and showed that the air collectors have a market share of less than 1% of the worldwide solar collector market. Even though PVT collectors are able to produce more energy compared to standalone system, the market share for such system is still negligible due to lack of proper public awareness. However, this situation should change since several institutes and manufacturers are making efforts to promote these systems. He also discussed the perspective for future development of a PVT collector and BIPVT system. It clearly shows that, by appropriate architectural design and configuration, the future of a PVT collector can be encouraging as an alternative application for the residential, industrial and commercial buildings.

In conclusion, Advantages of using air for cooling of PV cells are: 1 – Easily available and free of cost; 2 – Leakage is not dangerous and 3 – Inert fluid so no major reaction with any component material.

Based on the literature review presented above, air-based PV/T type performance is largely dependent on the air flow speed and temperature. The general trend in this research field is:

(i) Studying the more favorite air flow patterns,
(ii) Determining the optimized channel geometry and sizes to enable creating effective turbulence within the channels,
(iii) Selecting proper glazing modes e.g. uncovered or covering with single/double glass.
(iv) Development of standardized solutions for the integration of PVT air modules, possibly in combination with booster air collectors, in commercial air heating systems.

The major problem with the air-based system lies in its relatively poor heat removal effectiveness owing to the low density, specific heat capacity and thermal conductivity of the air.

With respect to systems level, the most important conclusion is that a serious lack in experience exists regarding systems performance, life time and reliability of PVT systems, due to lack of field tests and long term monitoring. It is very important that such projects are carried out, to generate experience, necessary for the development of standards for reliability and performance measurement, as well as for the generation of public confidence.

5. Economical assessment and applications of PVT collectors

The analysis presented until now concerned with characterization of the performance of hybrid systems in terms of energy and exergy efficiencies. Conclusions on the economic advantages of using hybrid collectors have not already been particularly addressed. An economic assessment is conditioned by the application context: the local weather conditions and the demands on heat to electric power ratio by the PVT system; limited economic development and policy matters. Nevertheless, an economic approach must always be preceded by an energy analysis of the system. Moreover, as the results that are involved in this analysis do not refer to instantaneous or steady state conditions, in order to achieve a comprehensive analysis it should be based on a long term assessment. Thus, simulation tools that include detailed local weather data and provide dynamic response of the model are necessary to carry out this type of analysis.

Ricaud and Roubeau (1994) developed an economic study of the combined generation of solar electricity and hot air in a residential and a commercial building. It was found that use of the considered hybrid system for the domestic application is in the limit of being competitive. Different metrics can be used for the economic evaluation of the application of hybrid PVT collectors. One of the most common is the *payback time* (PBT) of the system that can be determined by considering installation and operating costs, energy prices and economic figures, such as financial costs of loans, or inflation rate. The *energy payback time* (EPBT) is the total time period needed to recover the total energy spent to prepare the materials (embodied energy) used for fabrication of the systems. A life cycle analysis can also be carried out in order to obtain the total cost (or *life cycle cost*) and the *life cycle savings* (LCS) of the systems. This approach is followed in numerous studies (Kalogirou and Papamarcou, 2000; Brogren and Karlsson, 2001; Erdil et al., 2008; Chow et al., 2007). With the *Lyfe Cycle Savings* (LCS) method all costs and benefits are accounted for as their present values. General conclusions are not necessarily relevant, since are limited to a certain geographic application. However, some methodologies can be highlighted.

Kalogirou and Tripanagnostopoulos (2006) studied PVT domestic hot water solar systems using a-Si and pc-Si modules in three locations at different latitudes (Nicosia (35°), Athens (38°) and Madison (43°)). The economic analysis indicated better results of *LCS* for lower latitudes and a-Si modules. A similar study was

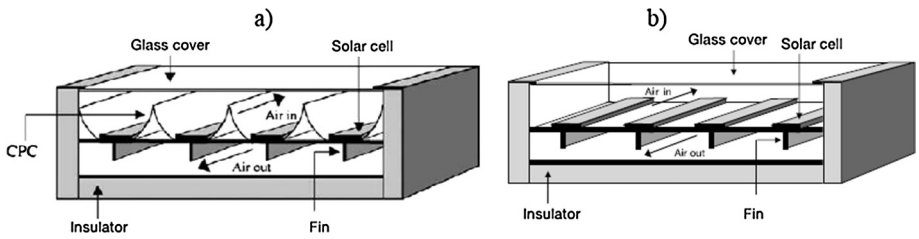


Fig. 11. Double pass PVT air heater improved configurations: (a) with CPC and fins (Othman et al., 2005); (b) with fins Othman et al., 2007.

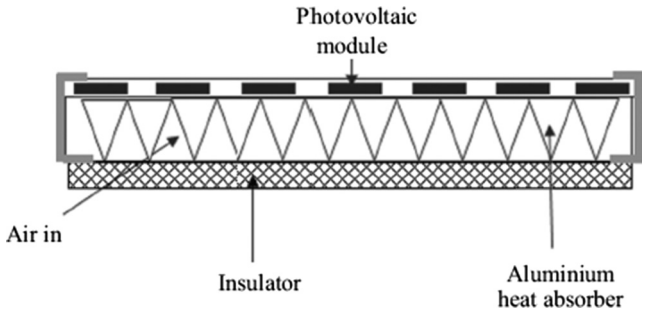


Fig. 12. PVT system with aluminum ∇-grooved absorber plate (Hussain, 2013).

expanded to include industrial applications (Kalogirou and Tripanagnostopoulos, 2007). Aste et al. (2012a, 2012b, 2015), calculated the optimal value of solar fraction (*fS*) for hybrid PV/T-water systems, from energetic end economic point of views. Mishra and Tiwari (2013), evaluated and compared the energy matrices of a hybrid PVT water collector applied in New Delhi under constant collection temperature mode with five different types of PV modules: c-Si, p-Si, a-Si (thin film), CdTe and CIGS. The highest annual overall thermal energy and exergy were obtained with the use of c-Si PV module. The maximum and minimum EPBT of 1.01 and 0.66 years was obtained for c-Si and CIGS respectively, whereas on exergy basis maximum EPBT of 5.72 years was obtained for a-Si and minimum of 3.44 for CIGS PV module.

The concept of payback time was applied in an environmental perspective by Tripanagnostopoulos et al. (2005), in a study on the annual performance of water cooled PVT solar systems, combining different configurations (glazed/unglazed, horizontal/tilted, different operating temperatures). The study included the estimation of costs and payback time, and determined the *energy payback time* (EPBT) and *CO_2 payback time*. When performing an economic assessment of hybrid systems, thermal and electrical outputs are accounted for separately, since they have different costs and energy savings.

An approach focusing on methods to develop a ratio between electrical and thermal output from a domestic PVT system was conducted by Coventry and Lovegrove (2003). Methods discussed included thermodynamic analysis using exergy, market analysis for both an open market and a renewable energy market, and environmental analysis using savings in greenhouse gas emissions. A discounted cash flow method was used to determine levelized energy costs from both electrical and thermal renewable energy sources, giving a ratio between electrical to thermal value of 4.24. Levelized energy cost was plotted against the energy value ratio to compare different options, like using a-si or c-Si cells, or identifying critical points. System performance was simulated using TRNSYS, and the calculations were performed using US financial data.

In order to evaluate the economic viability of water heating PVT systems installed in three cities of different longitudes and climatic conditions (Athens, Munich and Dundee), Axaopoulos and Fylladitakis (2013), performed a simple economic analysis based on the LCS method. Since most of the economic parameters change in relation to time and geographic area, it is difficult to make reliable predictions about future trends in the value of money. Therefore, a sensitivity analysis, based on the *Net Present Value* (NPV) method was carried out to evaluate the economics of energy produced under various investment costs, feed-in tariff prices and energy price fluctuations. In all three areas and with the energy prices of that period, the installation of a PVT system seems to be a more advantageous investment when the auxiliary water heating energy source comes from electricity, followed by heating oil and then natural gas. The price of the thermal energy delivered by a water PVT collector was calculated by Evola and Marletta (2014) to be ranging from 1 to 10 c€/kW h, for an assumed price of the electricity ranging from 0.18 to 0.24 €/kW h.

In conclusion, there are different approaches to the analysis of economic performance of hybrid systems. Economical techniques are just one tool that can be used to describe situations for an advantageous application of PVT systems.

5.1. Space applications of PVT collectors

The most typical applications of hybrid collectors, a water heating and BIPVT systems combined with production of electricity. However, PVT technology presents good potential for other important applications. Its use can be a particular success for reducing the operation cost of space heating and cooling units, through solar heat pumps integrated with air/conditioning. PVT concepts can also be applied in solar green-houses and solar stills. Sometimes the innovation may not mean innovative or cutting-edge expensive technologies, but taking advantage of the right combination of the strengths of each one for some particular purpose. Van Helden et al. (2004) highlighted some possible examples. The available roof area, the demands of electricity and heat, and temperature levels determine the PVT system set-up that is most suited for a given house or building. For instance, an array of PVT collectors can be connected to an earth-coupled heat exchanger, so that the low temperature heat gained in the summer is kept in the ground and used in the winter. PV modules can provide the electricity to the heat pump that transfers energy from a low-temperature heat source to higher temperatures sink for either room heating or for DHW. This strategy can highly improve the solar fraction and over all efficiency of solar systems.

5.2. Heat pump systems

The heat pump coefficient performance (COP) is used to evaluate the heat pump efficiency, defined as the ratio of the amount of energy provided by the heat pump (Q) to electrical energy consumed. The COP of a heat pump becomes better with increasing evaporating temperature (fixed condensing temperature). The integration of a PVT system with a heat pump provides both thermal energy and electrical power to run the cycle, e.g., for space heating. The common potential benefit to the solar collector and the heat pump performance is to depress the collector temperature and boost the heat pump evaporator temperature, enhancing the performance of both systems, and thus improving the heat pump COP. The integrated system is called Solar Assisted Heat Pump (SAHP). If properly incorporated, SAHP systems can cover the

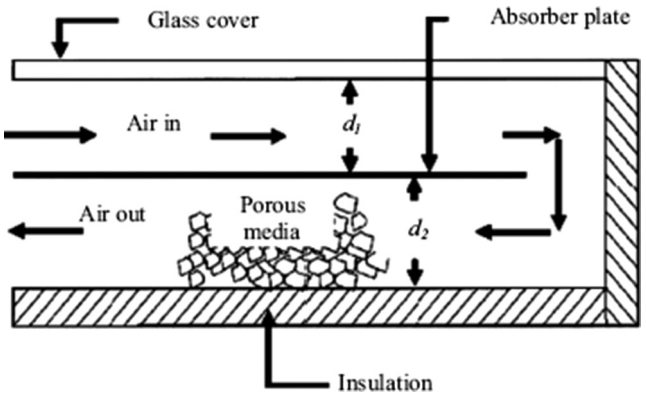


Fig. 13. Double pass PVT system with porous media.

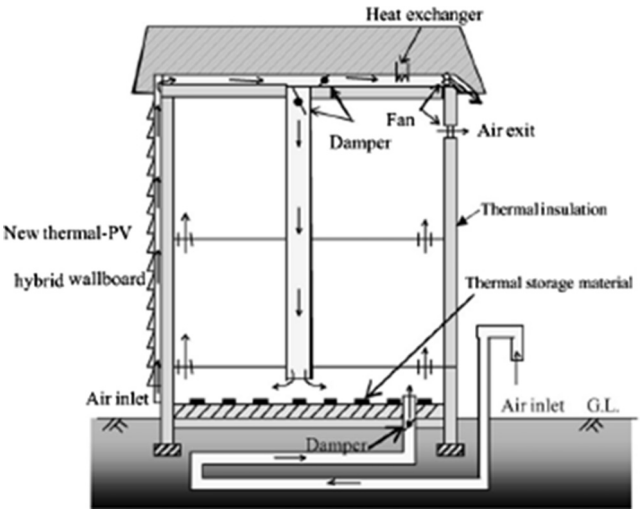


Fig. 14. System concept of hybrid wallboard with air (Nagano et al., 2003).

heating load in cold climates. A solar system could deliver a relatively high temperature source for a heat pump compared to a traditional heat pump using outdoor air and ground source heat exchanger (Kamel et al., 2015).

Solar assisted heat pumps are categorized, based on the integration scheme between the solar system and the heat pump, into *Direct Expansion Solar Assisted Heat Pump* (DX-SAHP) and *Indirect Expansion Solar Assisted Heat Pump* (IDX-SAHP). Most research and development of solar assisted heat pump systems consist of a solar thermal collector, which produces only thermal energy, linked with the evaporator of a heat pump, i.e., thermal energy is extracted from the solar thermal system for heating purposes only. The only integrated PVT system with a heat pump, which provides both thermal energy and electrical power, is the PV-evaporator type (Kamel et al., 2015).

In DX-SAHP, heat pump and solar system work together as one combined system. The PVT collector in the DX-SAHP system works as the evaporator low temperature heat source of the heat pump. A schematic diagram of the working principle of a direct expansion system is represented in Fig. 13. In this arrangement, liquid refrigerant vaporizes at the tubing underneath the flat plate collector. Solar energy is absorbed at the PVT evaporator that operates at a lower temperature than the ambient environment, and released later at the water-cooled condenser at a higher temperature. Cell efficiency is then higher than the standard operating efficiency. Take into account this working principle, a PV/T–SAHP system with pc-Si aluminum roll-bond panels was constructed and studied by Ito et al. (1997, 1999). The experimental results indicated that the COP of the heat pump is able to reach 6.0, and with hot water supplying to the condenser at 40 °C. A DX-SAHP system was also

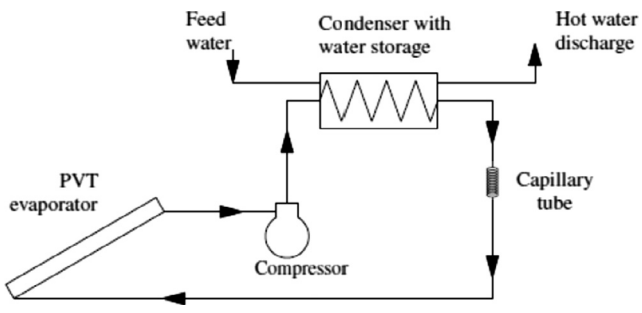


Fig. 15. Schematic diagram of the DX-SAHP system (Chow, 2010).

proposed by Chow (2010). Numerical analysis was carried out making use of a dynamic simulation model developed for the Hong Kong climate. It was found that the proposed system, working with R-134a, is able to achieve a yearly-average COP of 5.93 and PV output efficiency of 12.1%, thus with an energy output considerably higher than the conventional heat pump plus PV "side-by-side" system. The results for the monthly average COP for DX-SAHP using solar thermal collector and a PV-SAHP system, using a PVT collector, are shown in Fig. 14.

COP_{pt} definition referenced in the figure corresponds to:

$$COP_{pt} = \frac{Q + E/\eta_p}{W}$$

where Q is the heat transferred in the condenser, E is the output power of the solar cells per unit area, and η_p is the average electricity generation efficiency of the power plant as a conversion factor (= 0.38).

During the day, there is a chance possibility of liquid refrigerant remaining at the outlet of the PV-evaporator because solar radiation changes; consequently, the system works inefficiently and will require control of the mass flow rate of the refrigerant. In addition, the system works only when the evaporator temperature is above a certain value. It may be useful to overcome this issue by separating the PVT unit from the heat pump evaporator. In indirect expansion heat pumps, the heat pump and the solar collector are combined together, but work as two distinctive systems that can be arranged in parallel, in series, or dually. Air and water PVT collectors can be used in SAHP systems. Kamel et al. (2015), presents a review on the application of hybrid PVT to heat pumps systems. A summary of studies referring to IDX-SAHP systems and the specific application of SAHP for cold climates is pointed out.

5.3. Solar cooling with PVT technology

In climates with high radiation and extreme outdoor temperatures, there is a need to lower the indoor temperature considerably in order to provide thermal comfort. Solar cooling of buildings is part of the most attractive solutions. This is an application in which the demand for cooling energy closely matches the availability of solar energy, both in the seasonal and daily variations. One of the technologies of solar cooling is based on the absorption cycle, depicted in Fig. 15. Absorption is the process of attracting and holding moisture in substances called desiccants. Absorption systems are the same as vapor-compression air conditioning systems but differ in the pressurization stages. In general an absorbent on the low-pressure side absorbs an evaporating refrigerant. The most usual combinations of fluids include lithium bromide–water (LiBr–H_2O), where water vapor is the refrigerant, and ammonia–water (NH_3–H_2O) systems where ammonia is the refrigerant. The pressurization is achieved by dissolving the refrigerant in the absorbent in an absorber section. Subsequently, the solution is pumped to a high pressure with an ordinary liquid pump. The addition of heat in the generator is used to separate the low-boiling refrigerant from the solution. In this way the refrigerant vapor is compressed without the need for large amounts of mechanical energy that a vapor-compression air conditioning system demands (Florides et al., 2002). Therefore, the absorption chillers can be powered by solar collectors. The COP for conventional absorption cycles is about 0.6–0.7 for NH_3–H_2O systems (Florides et al., 2002), and between 0.6 and 0.8 for LiBr–H_2O systems (Duffie, 1991). This solution, however, needs electricity to drive the pump, despite the power is very small (see Fig. 16 and Fig. 17).

Simultaneous production of electrical and high-grade thermal energy is proposed with a concentrating photovoltaic/thermal (CPVT) system operating at elevated temperature by Mittelman

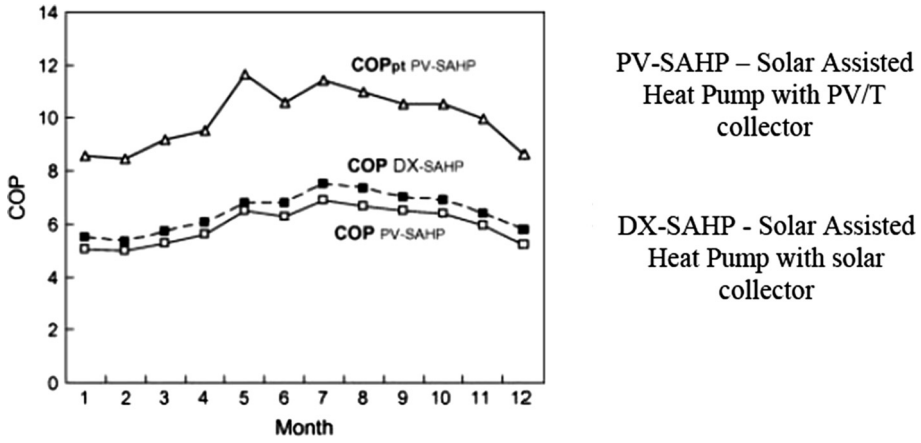


Fig. 16. Comparison of monthly average COP between traditional DX-SAHP and PVT SAHP (Chow, 2010).

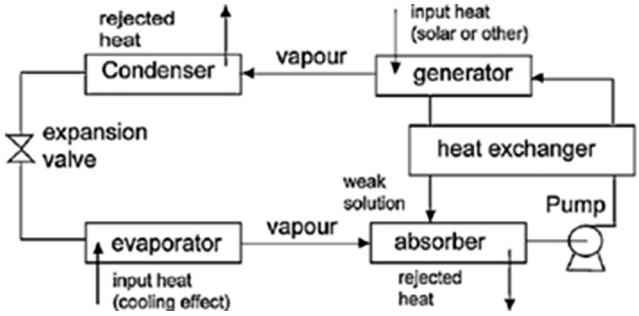


Fig. 17. Basic principle of the absorption air conditioning system (Florides et al., 2002).

et al. (2007). CPVT collectors may operate at temperatures above 100 °C, and the thermal energy can drive processes such as refrigeration, desalination and steam production. The performance and cost of a CPVT system with single effect absorption cooling were investigated in detail. An example of a thermal application that can be combined to the CPVT collectors, a single effect absorption chiller, was analyzed under different scenarios. The results demonstrate that under a reasonably wide range of conditions, the CPVT cooling system can be comparable in costs to a conventional alternative. Under some conditions, the solar cooling is even significantly less costly than conventional cooling. This is in contrast with solar cooling based on thermal collectors, which is usually found to be considerably more expensive than conventional cooling. The range of conditions where CPVT cooling is viable depends on the prevailing local costs of conventional energy. Nanotechnology also made it possible to use conventional Photovoltaic (PV) system as Photovoltaic Thermal (CPVT). In recent years, nanofluids have been proposed as efficient coolant fluids and as a means to filter sunlight for PV/T systems (Jing et al., 2015).

Recently, a three dimensional concentrated PV/T model was built and its photovoltaic/thermal system performances were investigated by a coupled TRNSYS and CFD simulation method by Li et al. (2017).The influences of different kinds of solar cells and various parameters on the system performance have been systematically evaluated. As the most important economic index for the practical use of PV/T system, in terms of the overall exergy efficiency under 50 times concentrations and 0.08 m/s inlet velocity, It was analyzed per hour, per month and per day based on the meteorological condition of Xi'an, a typical city in west China. Results found that the optimum working conditions vary according to different evaluation factors, the reasonable results should be determined according to specific needs.

5.4. Tri-generation and poly-generation application

As new applications for hybrid collectors are studied, new possibilities for the integration of diverse technologies arise. A further possible use of both medium-temperature and high temperature PVT collectors is the integration with solar heating and cooling (SHC) technology. SHC systems can utilize solar radiation to provide space heating during the winter and space cooling in the summer, by using a heat-driven chiller, which can either work by absorption or adsorption process (Calise et al., 2017). The heat required for that process is provided by a solar collector field. The use of PVT collectors, instead of solar thermal ones, can provide electricity additionally. An energy supply system which delivers simultaneously more than one form of energy to the final user is the basis of the poly-generation concept. The association of cooling, heating and power in one system is also called Combined Cooling, Heat and Power (CCHP), or tri-generation.

A SHC system with photovoltaic/thermal collectors was studied by Calise et al. (2012) applied to a university building located in Naples. The system integrated PVT collectors, a single-stage LiBr–H_2O absorption chiller, storage tanks and auxiliary heaters. Electricity generated through PVT collectors was partially consumed in the building, by electric appliances, lighting and system parasitic loads. The surplus was marketed to the grid. Simultaneously, the PVT system provides the heat needed to drive the absorption chiller. The system performance was analyzed from both energetic and economic perspectives, using a transient simulation model developed in TRNSYS. The economic results showed that the system under investigation could be profitable, provided that an appropriate funding policy is available. In addition, overall energetic and economic results are comparable to those reported in literature for similar systems.

A coupled system is proposed by Mittelman et al. (2009) comprised of a concentrating PVT collector field and a multi-effect evaporation desalination plant. A desalination process utilizing solar energy is a logical combination, since regions with abundant solar radiation are often also short in potable water supply. Three processes are commercially available for large-scale desalination plants: Reverse Osmosis (RO), Multi-Stage Flash distillation (MSF), and Multiple Effect Evaporation (MEE). RO requires electricity, a high-grade form of energy; however, MEE and MSF consume thermal energy, and can readily operate with alternative low-grade heat sources such as solar energy. PVT collectors can thus be used

also for this application, with such a great importance for sustainability. However, large scale of desalination plants uses high inlet temperatures are required, in order to provide reasonable performance and reasonable heat exchanger sizes. These higher temperatures can be achieved with a concentrating collector field. The combined system studied produces solar electricity and simultaneously exploits the waste heat of the photovoltaic cells to desalinate water. A detailed simulation was conducted to compute the annual production of electricity and water. The cost of desalinated water was estimated and compared to that of alternative conventional and solar desalination plants, under several economic scenarios. The results indicate that the proposed coupling plant can have a significant advantage relative to other solar desalination approaches. For a scenario with higher electricity prices, CPVT desalination is even more cost-effective than conventional desalination with consequences of zero or even negative cost of the desalinated water. Calise et al. (2014) investigated the integration of renewable energy sources and water systems, presenting a novel solar PVT system producing simultaneously electrical energy, thermal energy, cooling energy and desalinized water. The system is designed for small communities in European Mediterranean countries, rich in renewable sources and deficient in fossil fuels and water resources. It includes PVT collectors, a MED system for seawater desalination, a single-stage LiBr–H_2O absorption chiller and additional components, such as storage tanks, auxiliary biomass-fired heaters and devices for the balance of plant. Since absorption chillers and MED process require for high temperature water, it is needed the use of concentrating PVT collectors. The system is dynamically simulated by means of a zero-dimensional transient simulation model. A thermo-economic analysis is also presented, aiming at determining the optimal values of the most important design variables, making use of TRNSYS. A numerical case study was developed and widely discussed, putting in evidence the expressive potential of energy savings achievable by such system, also due to the opportunity of maximizing the utilization factor of the thermal energy produced by the CPVT, especially during the summer. On the other hand, the winter performance was by far less satisfactory, since CPVT thermal and electrical productions dramatically decrease; as a consequence, in such periods a large amount of heat must be generated by an auxiliary heater to drive the MED.

5.5. Solar stills and solar greenhouses

Solar stills are small-scale, decentralized, environmentally friendly solution for getting pure water through desalination. However, conventional single basin passive solar stills is not widely used because of their low yield (approximately 2–3 l/m^2 day), low thermal efficiency (maximum around 30%) dependent on solar intensity which varies with location. A hybrid (PV/T) active solar still is a combination of solar still and flat plate collector integrated with glass–glass photovoltaic module. A PVT solar still was proposed by Kumar and Tiwari (2009). An analysis of the annual performance and cost of distilled water produced was carried out and compared with passive solar stills for the climate of India. The comparative cost of distilled water produced from passive solar still was found to be less than hybrid (PV/T) active solar still for 30 years lifetime of the systems. The payback periods of the passive and PVT active solar still are estimated to be in the range of 1.1–6.2 years and 3.3–23.9 years, respectively. Despite the higher price, the use of PVT solar still can be advantageous when electricity is unavailable. The main objective of the work of Singh (2011) was enhancing the productivity of a double slope solar still to provide distilled water for isolated communities, facing electricity problems and good quality of water for commercial use. To increase the temperature of the feed water in the solar still, a double slope solar still was connected to two flat plate collectors with one of their photovoltaic (PV) integrated. It was found that energy payback time is significantly reduced by almost 30% in present design with less capital investment.

Energy and exergy performance analysis prediction of PVT collector integrated with a greenhouse at I.I.T, Delhi in India was conducted by Nayak and Tiwari (2008). The analysis is based on quasi-steady state conditions. Experiments for the annual performance and numerical computation were carried out for a typical day only for validation. Exergy analysis calculations of the PVT integrated greenhouse system showed an exergy efficiency level of approximately 4%. Sonneveld et al. (2009) presented a greenhouse with a spectral selective coating on the covering. This coating reflects near infrared radiation (NIR), containing about 50% of the solar heat load which is not required for crop production, and transmits the PAR. Because the NIR reflection is performed with a bent surface, the whole covering worked as a solar concentrator PV module located in the focal line. This concentrator technology reduces the surface area required by the expensive PV cells, so only a small PV area is needed. The module is cooled with water so the system delivers both electrical and thermal energy. Further performance results on this type of systems were presented by Sonneveld et al. (2010). After the description of the construction of this greenhouse, the peak power for Dutch weather circumstances is determined based on the amount of electrical and thermal energy (hot water) produced. The typical yearly yield of this greenhouse system is determined as total electrical energy of about 20 kW h/m^2 and thermal energy of 160 kW h/m^2.

5.6. Applications of hybrid collectors in buildings

Liquid photovoltaic/thermal (PV/T) collectors are used to heat water and simultaneously produce electricity for various domestic and industrial applications (Kumar et al., 2015). The largest market potential is for domestic hot water (DHW) applications, possibly combined with space heating. Although most collectors are installed on single-family houses, the share of large systems for collective applications is expected to increase. The application of hybrid collectors in the industry is also an option, once they can partly satisfy the high demand of energy for both heat and electricity in industries. However, the temperature of the cooling medium must not be permitted to be too high as it will have a detrimental effect on the module efficiency. Hence, the system is inherently a low temperature set-up and will only serve applications where a pre-heated medium is preferably used (Erdil et al., 2008).

Zhang et al. (2012) refer that the studies related to economic and environmental analyses of the systems with hybrid collectors done so far are adequate to indicate the performance of the PVT technology in terms of its economic and carbon benefits, and addressing the following approaches:

(1) PVT energy saving potential, its cost augment, estimated payback time and life cycle cost saving;
(2) PVT Energy Payback Time and Greenhouse-gas Payback time and their relevance with the system's energy and exergy efficiencies.

However, not sufficient studies analyze the influence of the choice of P_f on system energetic, economic and environmental performances. This is an important issue for consumers (and also manufacturers), when deciding the best configuration for the hybrid PVT collector.

The assessment of the suitability of the different PVT technologies for use in buildings was performed using four indicators as presented in Table 3.

Table 3
Results of the evaluation of the suitability for use in building of different PVT technologies on four indicators (Wu et al., 2016).

Indicator/Technology	Ventilated PV	Air-based PVT	PVT liquid (covered)	PVT liquid (Uncovered)	Concentrating PVT
Building integration potential	*High*	*Medium*	*Medium*	*Medium*	*Low*
Electricity output.	*Increased*	*Increased*	*Decreased*	*Increased*	*N/A*
Thermal output.	*Indirect*	*Direct/indirect*	*Indirect*	*Indirect*	*Direct*
Available products.	*Low*	*Low*	*Low*	*Medium*	*Low*

It is known that PV/T technology has been used in many practical projects. However, there is little report found by literature search that focused on assessment of the system's performance under real climatic conditions. Research could be developed from this angle to develop, install and monitor the PV/T systems in real buildings. This will allow assessment of the real performance of the system including reliability and commercialization potential. Further, feasibility of the system used in real buildings could be examined.

In recent years, efforts have been employed with innovative passive cooling methods and techniques. Use of phase change material (PCM), in this regard, has been deemed to be a potential option by many researchers (Good, 2015). Regardless of the fact that many research works have been carried out on PV thermal regulation by employing PCM, its economic feasibility is still questionable. This problem along some other setbacks like unavailability of optimal PCMs that possess all favorable features for effective thermal control, lack of suitable disposal technology for the used PCM after completion of their life cycle, etc. held back the commercialization of PCM based photovoltaic systems. Nevertheless, there are great possibilities of this technology as PCM offer not only temperature reduction but also energy storage. Phase change materials have been investigated as a possible option to limit operating temperature in photovoltaic systems and increase efficiency. Compared with other options for temperature regulation such as forced air circulation design, a PV-PCM combination brings some positive features such as high heat absorption rate, no moving parts, no electricity consumption, and no maintenance cost (Huang et al., 2004). Hendricks and Sark (2013) in their review on the application of PCM in PV systems raised the point of commercialization of such hybridizations and analyzed the economic viability of PV-PCM systems. Browne (2015) presented such a study that provides a depiction energy economy of a PV-PCM system through thermal management and performance improvement. The authors used solid-liquid PCM in PV panels to normalize the operating temperature and conducted both electrical and thermal analysis to assess their effectiveness in two different climates of Dublin, Ireland and Vehari, Pakistan. Cost analysis reveals that hybrid PV-PCM systems are cost effective in the hot climatic conditions like that prevails in Vehari, Pakistan. But in a relatively cold environment like Dublin, Ireland this system is not that encouraging in term of financial benefit.

6. Performance assessment of systems with hybrid collectors

The benefits of PV/T systems are reaped most when there is a demand for both electricity and heat. It is observed that the electricity output of a PV/T collector is already greater than a similar sized PV collector. Adding the thermal output to that, having a PV/T system is most advantageous there is a demand for heat as well besides the increased electricity output. If there is a demand for heat only, having a solar thermal system would be more appropriate due to greater thermal performance and better economic feasibility of solar thermal systems. In case of electricity demand only, designing a PV system would be more suitable due to lower costs although PV/T systems offer better performance in electricity production (Allan et al., 2015).

The evaluation of the performance of PVT systems is complex and the comparison of the output energy is not a sufficient criterion to assess the performance of the PVT collectors, due to the interaction of thermal and electrical efficiencies and the different forms and values of the output energy.

As the thermal and electrical outputs depend on the incident radiation, the yield of solar collectors is not only determined by the quality of the collector, but at least as much by the weather and the type of system it is used in. For the effect of the climate, important differences are the annual amount of irradiance and the variation of the irradiance over the year. In particular, cold but sunny winter days are suitable for solar space/water heating. Due to the very high difference between summer and winter irradiance, the Northernmost climates require a large amount of storage to obtain a significant solar input for space heating, while regions such as the south of Canada, the USA or Japan can cover a much larger share of their heating load with direct solar heating (Zondag, 2008).

In terms of economic issues, simple payback time (Cost payback time, Energy payback time and Greenhouse payback time) and life cycle cost (LCC) were addressed taking into account primary fossil fuel energy saving and increase in capital cost and maintenance cost needed during the system operation. In terms of environmental issues, energy and exergy efficiencies of the system, Energy Payback Time and Greenhouse-gas Payback Time were calculated and used as the indexes to justify the benefits of the system in terms of the capacity of carbon emission cut.

Important parameters of system performance are the required temperature level and the solar fraction obtained. In thermal systems, the solar savings fraction, or solar fraction (f_S), is the useful energy obtained from solar energy divided by the total energy needs of a conventional system, without solar collectors. The solar fraction can be used to determine the Energy Savings, taking into account the energy consumption of the reference system (Hasan et al., 2014). The influence of the packing factor and the water flow-rate in the overall performance of hybrid PVT systems applied in a UK domestic building was considered in a study by Dupeyrat et al. (2014, 2009). The results show that, for the case of the UK (low solar irradiance and low ambient temperatures), a complete coverage of the solar collector with PV together with a low collector flow rate are beneficial in allowing the system to achieve a high coverage of the total annual energy (heat and power) demand, while maximizing the CO_2 emissions savings. In addition, the emission assessment indicates that a PVT system can save up to 16.0 tons of CO_2 over a lifetime of 20 years, which is significantly (36%) higher than the 11.8 tones of CO_2 saved with a PV-only system.

The application of the photovoltaic system is increasing in the market day by day. Although researches on new photovoltaic technologies are continually in progress and efficient PV cooling technologies are expected to disembark in the market. The lack of economic viability, public awareness, product standardization, warranties and performance certification, installation training, and experiences are the barriers. It is important for the reliability

of the technology to be thoroughly assessed. The improvements in power supply stability with power conditioner and better integration of renewable energy sources on to utility grid have been other key research areas. There are still some areas which need to be addressed by the researchers. An advanced research in the following areas is eagerly awaited. Although historically silicon (in the form of mono-, polycrystalline or amorphous) is the most used semiconductor material PV cell in the industry, there have been conducted some works to find better alternatives to silicon. But still there is a wide open area to carry on through research work to find such materials that would not suffer voltage (thereby power) drop under elevated temperatures. Research may also be carried out to find an appropriate PV module surface coating that will facilitate more effective radiative cooling.

In addition, use of phase change materials (PCM) may offer an optimum solution for commercial scale passive cooling. Though several works have carried out to find an efficient way of using PCM in PV cooling, considerable effort is required to encourage PV-PCM technology (Hasanuzzaman et al., 2016; Islam et al., 2016). Active cooling systems are very much capable ways for PV cooling even for solar farms, but the rate and amount heat dissipation is still a vital problem. Moreover, effective methods should be devised to make sure the utilization of the heat thus extracted. So, several extensive studies, both numerical and experimental, could be taken into account in order to find designs for heat exchangers that are suitable particularly for PV modules.

Works related to economic and environmental analyses cover:

(i) PV/T energy saving potential, its cost augment, estimated payback time and life cycle cost saving (Tripanagnostopoulos et al., 2005),
(ii) PV/T Energy Payback Time and Greenhouse-gas Payback Time and their relevance with the system's energy and exergy efficiencies,
(iii) Comparison among different PV/T configurations, PV alone, solar thermal alone and separately laid PV and solar thermal arrangements.

In summary, economic and environmental analyses works done so far are adequate to indicate the performance of the PV/T technology in terms of its economic and carbon benefits. The further work may be extended to long term (seasonal and annual) analyses of the system's performance by taking into account the influence of the climatic conditions to the system performance.

The technical-economic analysis is also a fundamental element in evaluating PVT technology efficiency. Technical-economic evaluation is a comparative study conducted which allowed choosing an installation of solar system by analyzing its thermal and electrical behavior. Bianchini et al. (2017) carried out the potential offered by photovoltaic/thermal (PV/T) solar system in Central Italy. The PV/T solar plant was remotely monitored and data on environmental conditions, module temperature, cooling fluid flow rate and temperatures and both electric and thermal power production were continuously acquired and stored. The results showed that PV cooling produces a negligible increasing in the electric yield (1–3%) and the PV/T solar system is able to reach a simultaneously generation of electricity (835 kW h/m^2) and heat (1600 kW h/m^2) at a mean outlet temperature over 40 °C. The paper also reports a techno-economic comparison of the PV/T solar plant with separate photovoltaic and flat plate solar thermal. Recently, Sahota and Tiwari (2017) present a review to explore assess of energy and economic performance of various PVT systems. Recent techniques to improve solar thermal systems have been highlighted and they observed that SPV integrated solar thermal systems are self-sustained and more economical in rural sectors.

The technical potential considers the limitations of the existing technology in exploiting a certain energy source. Among other things, it takes into account the geographical locations where the energy can be explored, technical limitations and the final efficiency of energy conversion. The economic potential weighs the price of exploiting the renewable energy source against the total energy produced and its cost. It also accounts for future benefits of exploiting renewable energy sources. These can be the decrease in greenhouse gases emissions, influence on the health and wellbeing of society and the value of energy supply independence. The technical potential and the economic potential must be evaluated in a dynamic way. Both these potentials may vary significantly with time. The same applies to the analysis of economic potential. The energy price that the renewable energy source replaces might change significantly in the near future. Also, factors like future introduction of direct costs per released quantity of greenhouse gases can be taken into account as mentioned before. When evaluated, we suggest that two objectives can be achieved:

1. Comparative analysis on the types of collectors used. For this purpose, a realistic environmental and external conditions by using weather and energy demand data as input is an important parameter. This will result in a realistic approximation of the annual yield of the combined system and will provide information on which option is optimal in economic terms and how this relates to a reference case. It is more meaningful to calculate the efficiency of the whole system, than for an individual PVT collector.
2. Identify the best choice for collectors and solar installation based on cost benefit criteria. In this case, it is important to maintain cost/benefits of the low-cost PVT collectors and to have a more precise estimation of all costs associated with the different systems (Carullo et al., 2017).

Also, despite very limited attention in literature, PVT systems reliability is a very important topic (Zondag, 2008). Major aspects that are believed to play an important role in the reliability of the system are:

– Stagnation temperatures: especially in glazed PVT collectors where high temperatures can be achieved (Van Helden et al., 2004) leading to the deterioration of the top film or encapsulation material.
– Thermal shock: rapid temperature changes due to a specific operation of the control scheme of the circulation pump could potentially be harmful for the lifetime of a PVT system.
– Ambient conditions: changing ambient conditions can result in high humidity, ice formation, hailstones, and other weather dependent consequences that could impact especially the exterior of the PVT system .

7. Summary

Efficient and economic harnessing of clean solar power may be very significant by fulfilling today's growing energy needs and tranquilizing climate concerns. It is expected in near future that solar energy will be economically competitive and it will be feasible to invest in and install solar systems even without subsidies and incentives. This paper provides an overview of recent studies of PVT thermal regulation techniques. The research in progress related to PV cooling technology is summarized above, to identify the unsettled problems and the prospects in this field. The various applications of the solar photovoltaic system such as building integrated air PVT system, solar air heating PVT system, the liquid PVT collector, concentrator PVT system, and heat-pipe-based PVT area are presented. Economic analysis using Levelized Cost of Energy

(LCE) for the laboratory scale set up for a non-cooled and cooled PV wherein the relative Levelized cost of energy was found to be reduced by 19.5%. Now various PVT applications on commercial level are there but still it is limited due to product reliability and cost. Hence significant research is required in the field of PVT mainly in thermal absorber design and fabrication, material and coating selection, energy conversion and its effective cost minimization, performance testing, control and their liability of the system. Some studies are until today had only been studied on a theoretical basis.

Although significant works have been completed in PV/T study, there are still some opportunities existing for further developing this technology, including:

(1) Developing new feasible, economic and energy efficient systems such as PCM-slurry-based PV/T,
(2) Optimizing the structural/geometrical parameters of the existing PV/T configurations,
(3) Studying long term dynamic performance of the PV/T systems,
(4) Demonstration of the PV/T systems in real buildings and feasibility study,
(5) Advanced economic and environmental analyses taking into account effect of the climatic conditions onto the performance of the system through long term measurement,
(6) Techno-economic analysis.

Further applications needed to be more studied such desalination and ply-generation systems. Also the use of heat pipes (with their different configuration) to PVT solar collector has not been extensively studied also PCM and thermoelectric systems can be another alternative in PVT technology where future focus area has also been proposed by several researchers. Although endeavors on the research and development work of PVT technology are in motion for the last 30–35 years, the economic aspects as well as some technical issues like energy storage problem are still to be addressed in wide ranging capacity. All these efforts aimed to create as much energy efficient PV/T system as possible at the least possible cost and simplest structure. This paper is found to be helpful for the manufactures of the PVT collectors, researchers and students who are working on this field.

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